Basic Network Sniffer

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Internship

Cyber Security Internship – Horizon TechX

Project Title

Basic Network Sniffer

Objective

The objective of this project is to understand how network packets travel across a network by capturing and analyzing live packets using Python and the Scapy library.

Tools Used

- Python 3.14
- Visual Studio Code
- Scapy Library
- Npcap
- Windows 10

Project Description

This project captures live network packets and displays useful information such as:

- Packet Number
- Source IP Address
- Destination IP Address
- Protocol (TCP/UDP/ICMP)

It helps in understanding basic networking concepts and packet analysis.

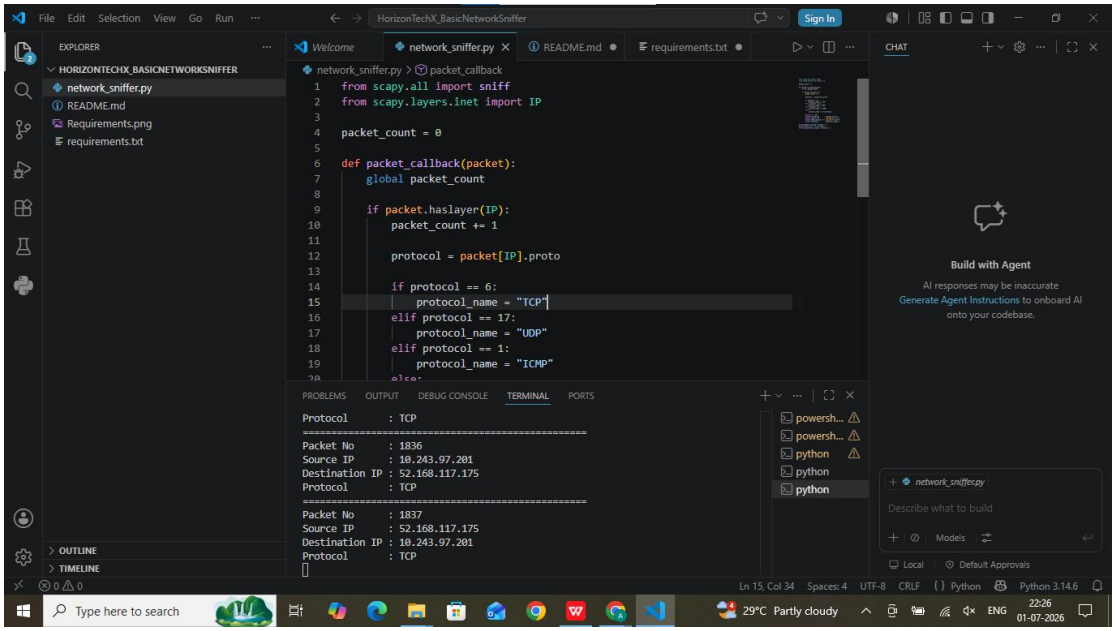
Output

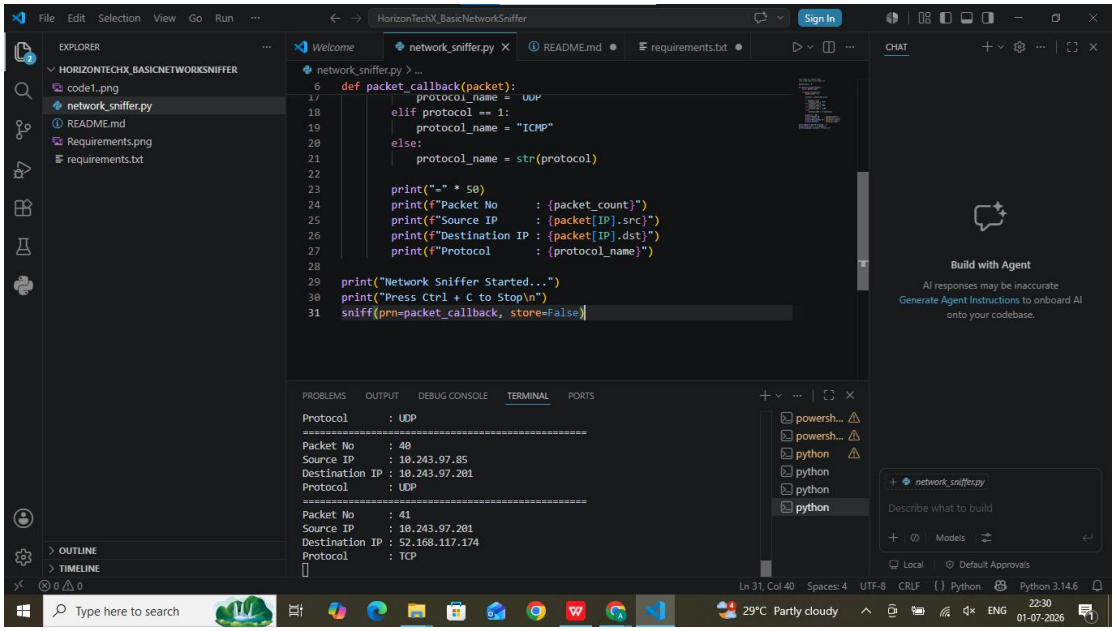
Paste your **output.png** screenshot here.

Conclusion

This project successfully captured live network packets and displayed Source IP, Destination IP, and Protocol information. It improved my understanding of network communication and packet analysis using Python.

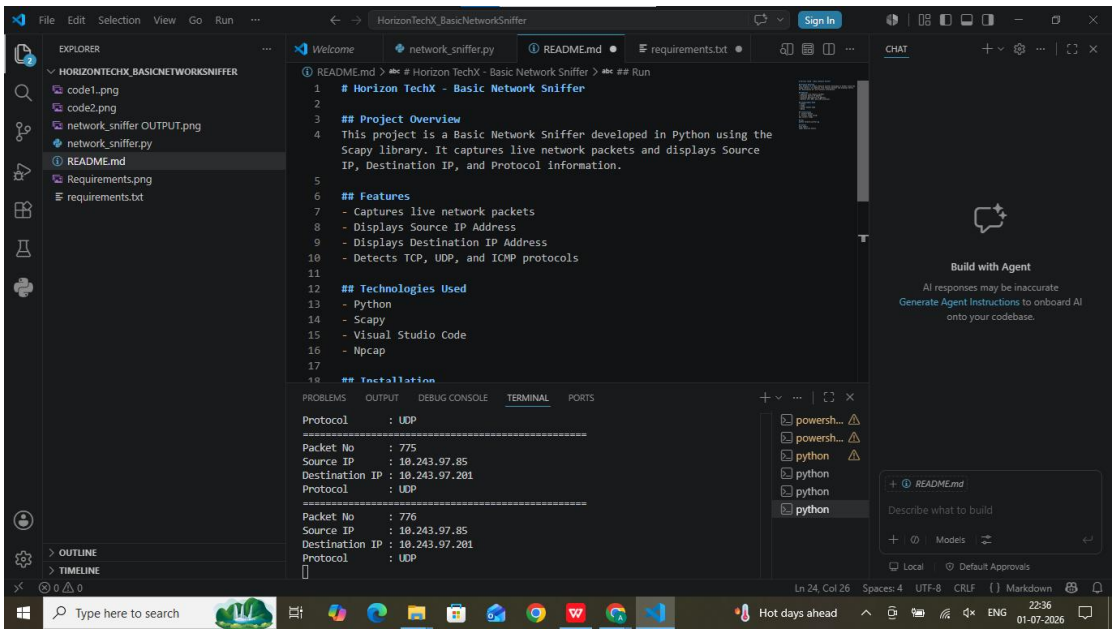
1. NETWORK SNIFFER

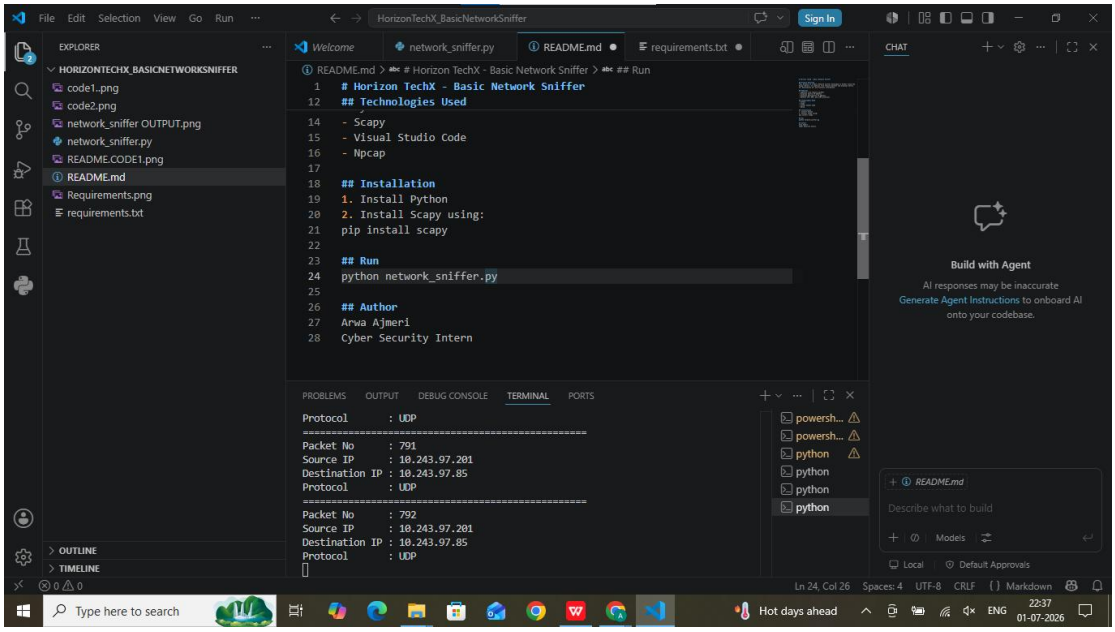


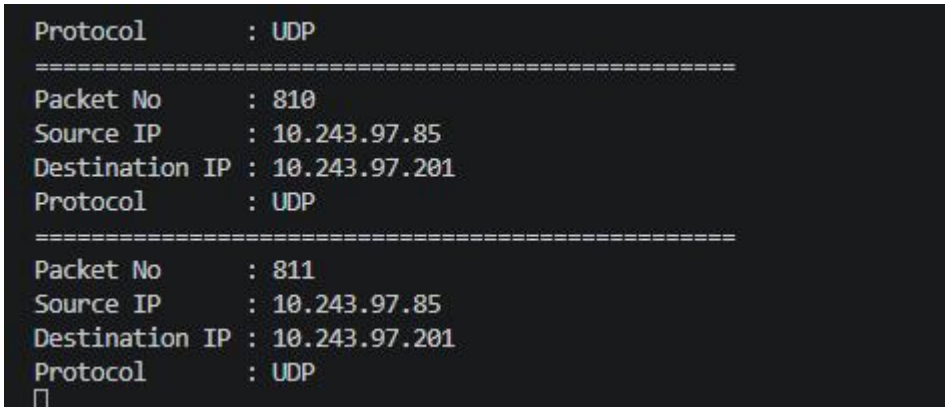


```
Protocol       : UDP
================================================
Packet No      : 47
Source IP      : 10.243.97.201
Destination IP : 10.243.97.255
Protocol       : UDP
================================================
Packet No      : 48
Source IP      : 10.243.97.201
Destination IP : 10.243.97.255
Protocol       : UDP
```

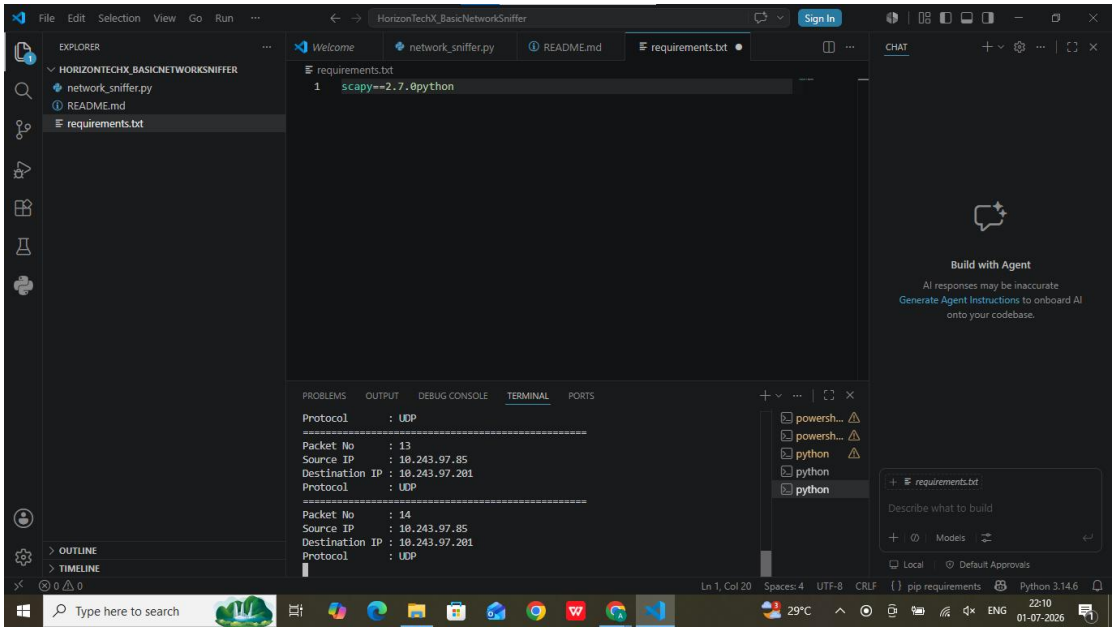
2. README.MD

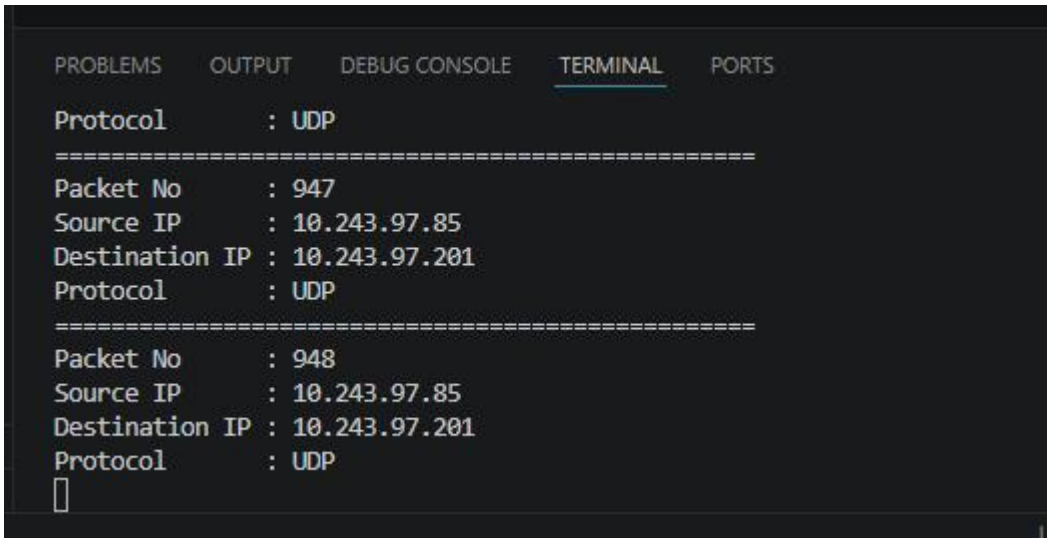






3. Requirements


File Edit Selection View Go Run
HorizonTechX_BasicNetworkSniffer
Sign In
EXPLORER
HORIZONTECHX_BASICNETWORKSNIFFER
network_sniffer.py
README.md
requirements.txt
Welcome
network_sniffer.py
README.md
requirements.txt
1 scapy==2.7.0python
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Protocol : UDP
Packet No : 13
Source IP : 10.243.97.85
Destination IP : 10.243.97.201
Protocol : UDP
Packet No : 14
Source IP : 10.243.97.85
Destination IP : 10.243.97.201
Protocol : UDP
powersh...
powersh...
python
python
python
Describe what to build
Models
Local
Default Approvals
OUTLINE
TIMELINE
Ln 1, Col 20
Spaces: 4
UTF-8
CRLF
pip requirements
Python 3.14.6
Type here to search
29°C
ENG
22:10
01-07-2026


PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Protocol : UDP
Packet No : 947
Source IP : 10.243.97.85
Destination IP : 10.243.97.201
Protocol : UDP
Packet No : 948
Source IP : 10.243.97.85
Destination IP : 10.243.97.201
Protocol : UDP